Lecture 12

Robustness to Unmeasured Confounders: Sensitivity Analysis

Dave Armstrong

Today's Robustness Question

What assumption/abstraction are we perturbing?

- All relevant confounders are measured and included in the model.

Why might this assumption fail?

- Relevant confounders could remain unmeasured or otherwise not included in the analysis.

Today's Robustness Question

What assumption/abstraction are we perturbing?

- All relevant confounders are measured and included in the model.

What kinds of inferential instability can result?

- Estimated treatment effects can be sensitive to potential correlated confounders.

Today's Robustness Question

What assumption/abstraction are we perturbing?

- All relevant confounders are measured and included in the model.

What tools exist to evaluate robustness?

- Sensitivity Analysis

Today's Robustness Question

What assumption/abstraction are we perturbing?

- All relevant confounders are measured and included in the model.

What are the limits of these tools?

- They can present a quite wide range of effects.
- Different methods provide different amounts of guidance about whether the required confounding is plausible.
- Some methods provide benchmarks using observed covariates, but all require assumptions that cannot themselves be verified.

Sensitivity Analysis

In general, sensitivity analysis is an effort to understand how our conclusions *could* change - If a covariate was added that related to our treatment and/or the outcome

The goal is not to "find the right answer", but to understand the conditions under which our answer might change.

- Generally, we will not *know* how likely such a scenario is, but there are some tools that help benchmark the result.

Prestige Model

```r
data(Prestige, package="carData")
# Prestige$income <- Prestige$income/1000
Prestige <- Prestige |>
  mutate(log_income = log(income),
         type_wc = as.numeric(type == "wc"),
         type_prof = as.numeric(type == "prof"))
mod <- lm(prestige ~ log_income + women +
            education + type_prof +
            type_wc, data=Prestige)
tidy(mod)
```

```
## # A tibble: 6 × 5
##   term        estimate std.error statistic       p.value
##   <chr>          <dbl>     <dbl>     <dbl>         <dbl>
## 1 (Intercept) -116.       18.8       -6.15 0.0000000196
## 2 log_income    14.7       2.31       6.34 0.00000000842
## 3 women          0.0838    0.0322     2.60 0.0108
## 4 education      2.97      0.602      4.94 0.00000349
## 5 type_prof      5.29      3.56       1.49 0.140
## 6 type_wc       -3.22      2.41      -1.34 0.185
```

What would it take to:

- Make the income effect insignificant?
- Flip the sign of the income effect?

Sensitivity analysis can answer these questions.

Sensitivity to Confounding Variables

Frank (2000) discusses sensitivity as robustness to the presence of unobserved confounders.

- Confounder: variable that correlates with both treatment and outcome (and is causally prior to the treatment).

$$y = \beta_0 + \beta_1 x + \varepsilon$$

β_1 can be causally interpreted if $\rho_{x,\varepsilon} = 0$ or alternatively $E(\varepsilon|x) = 0$.

ITCV: Impact Threshold for a Confounding Variable

- Amount of confounding required to make a significant effect cross the significance threshold.

Sensitivity

```
library(konfound)
k <- konfound(mod, "log_income", index="IT", to_return="table
```

```
## Dependent variable is prestige
##
## X represents log_income, Y represents prestige, v represen
## First table is based on unconditional correlations, second
## partial correlations.
```

```
k$Main_Output |> select(-statistic)
```

```
## # A tibble: 6 × 5
##   term        estimate std.error p.value   itcv
##   <chr>          <dbl>     <dbl>   <dbl>  <dbl>
## 1 (Intercept) -116.        18.8    0      NA
## 2 log_income    14.7        2.31   0       0.104
## 3 women          0.084      0.032  0.011   0.061
## 4 education      2.97       0.602  0       0.512
## 5 type_prof      5.29       3.56   0.14    0.489
## 6 type_wc       -3.22       2.41   0.185   0.044
```

ITCV measures *inferential robustness* - how big does a confounder effect have to be to change the inference I make?

- Not the same as "significantly" changing the effect size.
- an omitted variable would need partial correlations resembling education's to nullify the income result — still a real concern, but less alarming than the unconditional picture.

```
k$Partial_Impact |> select(1,4)
```

```
## # A tibble: 4 × 2
##   term       Partial_Impact
##   <chr>               <dbl>
## 1 women              0.147
## 2 education          0.218
## 3 type_prof          0.0159
## 4 type_wc           -0.0006
```

Robustness in Replacement

Whereas the ITCV is about potential *confounding variables*, the Robustness in Replacement (RiR) index answers

- how many observations would I have to replace with null effect observations to nullify the effect.

```
k2 <- konfound(mod, "log_income", index="RIR", to_return="raw
```

```
# Number of observations to nullify effect
k2$RIR_primary
```

```
## [1] 67
```

```
# percentage of observations to nullify effect
k2$RIR_perc
```

```
## [1] 68.67414
```

Cinelli and Hazlett (2020)

Frank (2000) tells us how much confounding is needed to push our result across the significance threshold.

- as Gelman & Stern remind us, the difference between significant and not significant is not necessarily itself significant.
- A confounder that moves our t-statistic from 6.3 to 1.9 has done something — but it hasn't shown the effect is zero.
- A confounder that moves the t-statistic from 2.5 to 1.5 may not be that problematic.

Cinelli & Hazlett answer a similar question, but provide more useful context

- how related would an omitted variable have to be to X and Y to change the significance.
- how related to change the sign?
- and how those requirements compare to the observed strength of covariates already in the model — giving a direct plausibility benchmark.

Three Sensitivity Statistics

Cinelli and Hazlett propose a sensitivity analysis framework that provides more concrete advice about what is a "good" and "bad" value (in the OLS context).

1. $R^2_{Y\sim D|\mathbf{X}}$ - the partial R^2 of Y with respect to the treatment controlling for $\mathbf{X}$.
   - A covariate that explained all residual variance in Y would have to explain at least 2.2% in the residual variance in the treatment to eliminate its effect.
2. RV - Unobserved confounders explaining at least $RV\%$ of the variance in both the treatment and the outcome could eliminate the effect of the treatment.
3. $RV_{\alpha=0.05}$ - Unobserved confounders explaining at least $RV_{\alpha=0.05}\%$ of the residual variance in both the outcome and treatment could render the treatment insignificant.

In R

```r
library(sensemakr)
s <- sensemakr(mod, "log_income")
s
```

```
## Sensitivity Analysis to Unobserved Confounding
##
## Model Formula: prestige ~ log_income + women + education +
##
## Null hypothesis: q = 1 and reduce = TRUE
##
## Unadjusted Estimates of ' log_income ':
##   Coef. estimate: 14.65518
##   Standard Error: 2.31151
##   t-value: 6.34009
##
## Sensitivity Statistics:
##   Partial R2 of treatment with outcome: 0.30407
##   Robustness Value, q = 1 : 0.4777
##   Robustness Value, q = 1 alpha = 0.05 : 0.36173
##
## For more information, check summary.
```

- What proportion of the residual variance of the treatment and control have to be explained by an unobserved confounder (orthogonal to the covariates) to move the coefficient across the significance threshold? **36.2%**
- What proportion of the residual variance of the treatment and control have to be explained by an unobserved confounder (orthogonal to the covariates) to move the coefficient across zero? **47.7%**
- What proportion of the residual variance of the treatment would an unobserved confounder that explains 100% of the outcome residual variance have to explain to fully account for the observed treatment effect? **30.4%**

Benchmark Variables

Benchmark variables allow us to evaluate whether unobserved confounders would need to be stronger than observed confounders to eliminate the effect.

1. $R^2_{Y\sim Z|D,\mathbf{X}}$ - the partial R^2 of Y with respect to the benchmark variable Z controlling for $\mathbf{X}$ and the t
   - If $R^2_{Y\sim Z|D,\mathbf{X}} < RV$, a confounder as strong as the benchmark would not be able to eliminate the treatment effect.
2. $R^2_{D\sim Z|\mathbf{X}}$ - the partial R^2 of the treatment with respect to Z.
   - If $R^2_{D\sim Z|\mathbf{X}} < R^2_{Y\sim D|\mathbf{X}}$, then even in the worst case secnario where an unobserved covariate explained all the residual variance in the outcome and is as strongly associated with the treatment as the benchmark covariate it would still not eliminate the effect of the treatment.

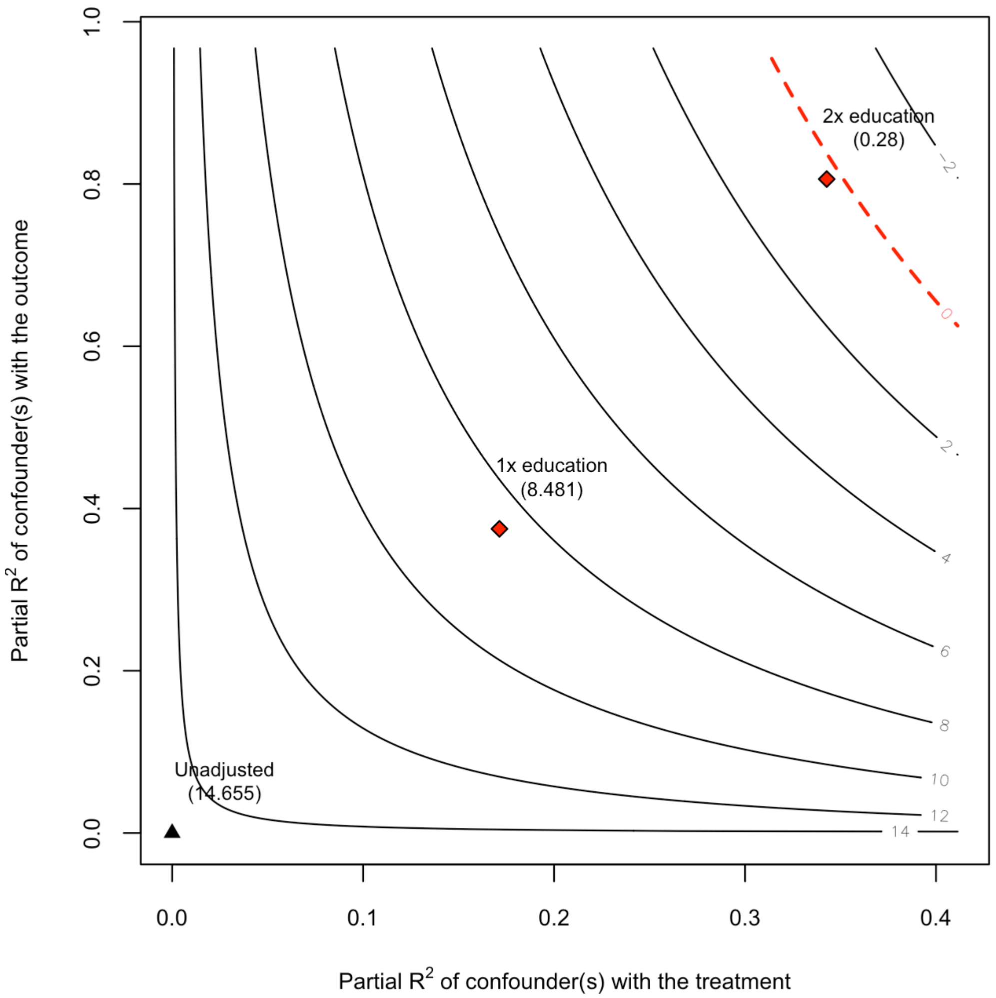
Benchmark Output

```
s <- sensemakr(mod, "log_income", "education", kd=1:2)
s$bounds
```

```
##    bound_label    r2dz.x    r2yz.dx  treatment adjusted_estimate adjusted_se
## 1 1x education 0.1713843 0.3749099 log_income         8.4812432    2.018664
## 2 2x education 0.3427685 0.8060845 log_income         0.2797287    1.262456
##   adjusted_t adjusted_lower_CI adjusted_upper_CI
## 1  4.2014142          4.472002         12.490484
## 2  0.2215749         -2.227619          2.787076
```

Plot

```
plot(s)
```



1.0
0.8
0.6
0.4
0.2
0.0
Partial R2 of confounder(s) with the outcome
0.0
0.1
0.2
0.3
0.4
Partial R2 of confounder(s) with the treatment
2x education
(0.28)
1x education
(8.481)
Unadjusted
(14.655)
-2
0
2
4
6
8
10
12
14

Oster (2019): Coefficient Stability and Selection

Frank and Cinelli & Hazlett ask: *how strong would a hypothetical omitted variable need to be?*

Oster asks a related but different question:

> **What does the behavior of your coefficient as controls are added tell you about omitted variable bias?**

The key insight: if adding observed controls moves your coefficient, that movement is informative about how much unobserved selection might remain — provided we're willing to make an assumption about the *relationship* between observed and unobserved selection.

The Proportional Selection Assumption

Let:

- $\dot{\beta}$, $\dot{R}^2$ = coefficient and R^2 **without** controls
- $\tilde{\beta}$, $\tilde{R}^2$ = coefficient and R^2 **with** controls
- R^2_{max} = the R^2 a model with *all* relevant variables would achieve

Oster's key assumption: selection on unobservables is proportional to selection on observables, scaled by δ:

$$\beta^*(\delta, R^2_{max}) \approx \tilde{\beta} - \delta \left[(\dot{\beta} - \tilde{\beta}) \times \frac{R^2_{max} - \tilde{R}^2}{\tilde{R}^2 - \dot{R}^2} \right]$$

- $\delta = 1$: unobservables generate as much selection as observables
- $\delta > 1$: unobservables generate *more* selection than observables

The δ^* Statistic

Solve for the δ that sets $\beta^* = 0$: $\delta^* = \frac{\tilde{\beta}(\tilde{R}^2 - \dot{R}^2)}{(\dot{\beta} - \tilde{\beta})(R^2_{max} - \tilde{R}^2)}$

Interpretation:

- $\delta^* > 1$: unobservables would need to generate *more* selection than observables to explain away the result — **robust**
- $\delta^* < 1$: unobservables need to generate *less* selection than observables — **fragile**
- $\delta^* = 1$: the benchmark of equal selection

Compared to Frank: Frank asks what correlation an omitted variable needs. Oster asks how much stronger unobserved selection needs to be than observed selection.

The R^2_{max} problem: δ^* is sensitive to the choice of R^2_{max}. Oster suggests $R^2_{max} = \min(1.3\tilde{R}^2, 1)$ as a default — but this is itself a sensitivity parameter.

Oster Statistics in R

```
o_delta(
  y = "prestige",
  x = "log_income",
  con = "women + education + type",
  type="lm",
  R2max = 0.9,
  data = Prestige
)
```

```
o_delta(
  y = "prestige",
  x = "log_income",
  con = "women + education + type",
  type="lm",
  R2max = 0.95,
  data = Prestige
)
```

```
## # A tibble: 7 × 2
##   Name                     Value
##   <chr>                    <dbl>
## 1 delta*                   1.60
## 2 Uncontrolled Coefficient 21.6
## 3 Controlled Coefficient   14.7
## 4 Uncontrolled R-square    0.549
## 5 Controlled R-square      0.865
## 6 Max R-square             0.9
## 7 beta hat                 0
```

```
## # A tibble: 7 × 2
##   Name                     Value
##   <chr>                    <dbl>
## 1 delta*                   1.04
## 2 Uncontrolled Coefficient 21.6
## 3 Controlled Coefficient   14.7
## 4 Uncontrolled R-square    0.549
## 5 Controlled R-square      0.865
## 6 Max R-square             0.95
## 7 beta hat                 0
```

Note: the sensitivity analysis is quite sensitive to R^2_{max}.

Clarke, Kenkel and Rueda (2018)

Clarke et al ask…

- From a sensitivity point of view, is adding more controls necessarily better?

Clarke, Kenkel and Rueda (2018)

Clarke et al ask…

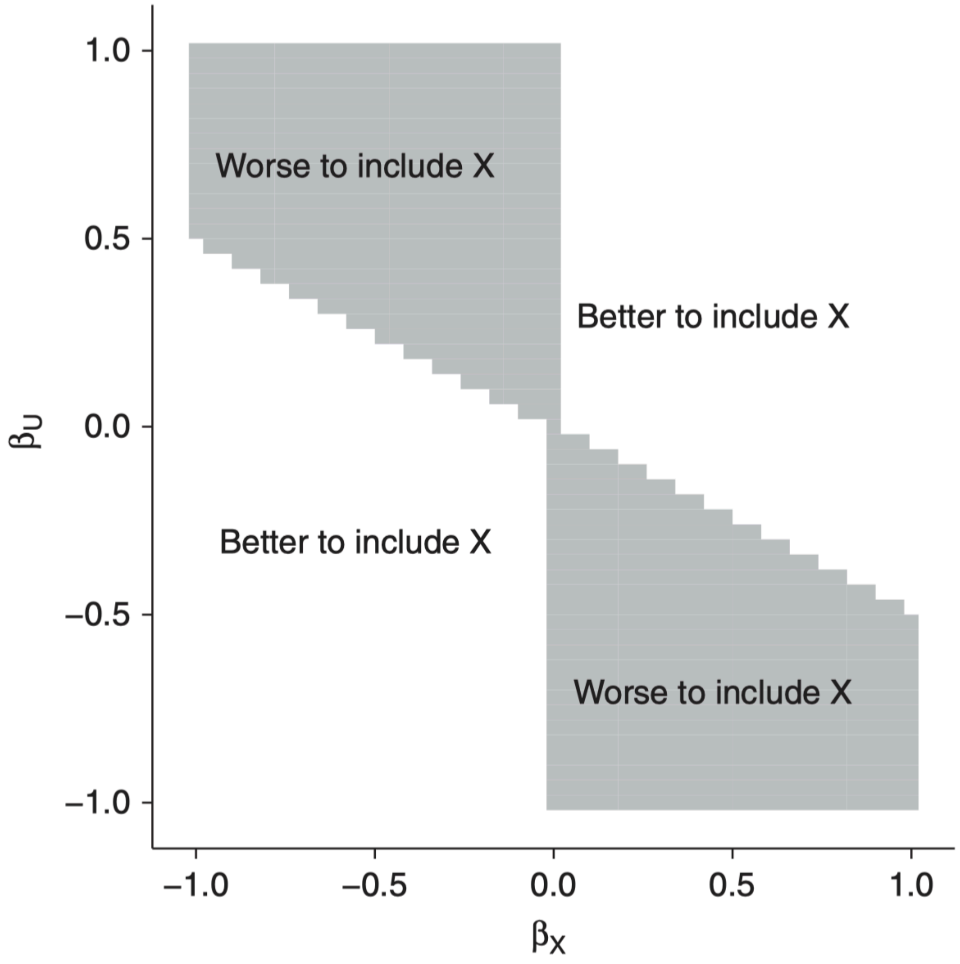
- From a sensitivity point of view, is adding more controls necessarily better?

Answer: Not necessarily

When observed and unobserved confounders pull the bias in opposite directions, controlling for the observed one removes its countervailing pressure — leaving bias of the unobserved confounder to act unopposed.

- You've made things worse by eliminating one side of a tug-of-war.

When is it better to condition on X?



With observed cofounder X and unobserved confounder U, it is *worse* to control for X if

1. U is a confounder *and*
2. U and X have countervailing effects *and*
3. The confounding effect of U is at least as great as that of X

Robustness Workflow

1. Estimate the preferred model.
2. Identify the treatment effect of interest.
3. Evaluate robustness using sensitivity analysis.
4. Benchmark against observed covariates.
5. Ask whether the required confounding is substantively plausible.
6. Interpret conclusions in light of the assumptions of the sensitivity method.

Remember:

- Sensitivity analysis does not tell us whether our estimate is correct.
- It tells us what assumptions would have to be false for our conclusion to change.